

IcomRigControl — What This App Can Do (Capabilities Inventory)

A complete, plain-language inventory of everything IcomRigControl does today, plus what's planned but not built. Meant as a shared reference and a brainstorming springboard. Grounded in the real code as of 2026-08-15 (Phases 1–10 complete, 283 passing tests).

In a nutshell: IcomRigControl is a free, cross-platform (Windows/macOS/Linux/Raspberry Pi) control and logging program for the Icom IC-7300 and IC-7300MK2 radios. It does everything Icom's own RS-BA1 remote software does *except* live remote audio — and adds a whole logging-and-operations ecosystem (QSO log, contests, callsign lookup, LoTW, HRD, N1MM/WSJT-X, APRS beaconing) that RS-BA1 has none of.

1. Radios & connection

Capability	Detail
Supported radios	Icom IC-7300 (CI-V address 0x94) and IC-7300MK2 (0xB6)
Local connection	USB / serial CI-V, selectable COM port / <code>/dev/tty...</code> , 115200 baud default
Remote connection	TCP to a networked CivTcpServer (Phase 9) — token-authenticated
Demo mode	Full UI with a simulated radio, no hardware needed
IC-705	Researched (address 0xA4, commands largely compatible) — not yet implemented

2. Core radio control

- **Frequency** — read live from the radio and set it (BCD-encoded CI-V, verified against the reference).
- **Mode** — LSB, USB, AM, CW, RTTY, FM, CW-R, RTTY-R, DV — read and set.
- **PTT (transmit) control** — key the radio on/off from the app (CI-V 1c 00), with the safety guarantees hardened in this session (never stuck on, never unidentified).
- **VFO / split** — VFO select, A=B, swap, split control commands are implemented in the engine.
- **Live meters, polled ~continuously:** S-meter (S-units + true dBm), RF power output %, SWR, ALC, supply voltage (Vd), current draw (Id).

3. Spectrum scope & waterfall (Phase 7)

- Live **spectrum scope** capture from the radio (27 xx commands).
- **Waterfall display** with a real **frequency axis** (labels in kHz/MHz).
- **Click-to-tune** — click a point on the waterfall and the radio tunes there.
- Configurable span.

4. Memory channel editor (Phase 4)

- **Bulk read** all 99 memory channels from the radio (skips empty ones), with progress.
- **Bulk write** channels back to the radio.
- Add/edit channels in a grid (frequency, mode).
- Cancelable long reads.

5. QSO logging — the resilient backup of record (Phase 8)

The core design principle: every contact lands in IcomRigControl's own local log, independent of any external program's health. Hardened this session so a QSO can never be lost to a write failure or a thread race.

- **Log a QSO** — auto-fills frequency, band, and mode from the live radio at the moment of logging.
- **Write-through persistence** — every QSO is immediately written to a timestamped ADIF session file (crash-safe).
- **ADIF export** — standard .adi output.
- **General + contest logging** in one UI, with a contest selector.
- **Contests supported (verified against official ARRL rules):** ARRL Field Day and ARRL RTTY Roundup — including exchange fields, dupe checking, serial numbers, and live score.

6. Callsign lookup (Phase 8c)

- Look up a callsign to auto-fill name and grid square while logging.
- **Sources:** QRZ.com (XML), HamQTH (XML), Callook.info (US) — user-selectable.
- Lookups are contractually non-throwing (never crash the logger); hardened with an extra guard this session.

7. Logbook of the World / LoTW (Phase 8d)

- **Upload** logged QSOs to ARRL LoTW (signing delegated to ARRL's own TQSL tool — never reimplemented).
- **Download confirmations** and mark confirmed QSOs (✓) in the log grid.

8. Ham Radio Deluxe integration (Phase 8e)

- One-way, best-effort bridge that writes QSOs into HRD Logbook's SQLite database (defensive, schema-checked). HRD being down never blocks the local log.

9. N1MM Logger+ / WSJT-X integration (Phase 8f)

- **Send** live rig state (frequency, mode, PTT) out as RadiInfo UDP packets, to one or more user-configured destinations (127.0.0.1 or explicit LAN).
- **Receive** logged contacts over UDP from N1MM/WSJT-X/HRD and fold them into the local log.

10. Activity logging (Phase 5)

- Continuous **CSV activity log** of rig state over time (timestamp, frequency, mode, all meter readings) for later analysis. Separate from the QSO log.

11. Remote / network operation (Phase 9)

- **Headless server mode** (`IcomRigControl.UI --headless-server`) — runs with no display, serves CI-V control to remote clients over TCP. Designed for a Raspberry Pi sitting next to the radio, reachable over LAN, VPN, or 44Net/AMPRNet.
- **Token authentication** required (never runs with a blank token).
- The desktop app can act as a **remote client** — from the app's perspective a remote radio is indistinguishable from a local one; every feature above works remotely.
- *Proven via loopback; real-hardware-over-real-network test still pending.*

12. APRS beacon over HF (Phase 10)

Neither IC-7300 has a TNC — this is built as software AFSK/AX.25 packet audio, complete and confirmed audible on both Windows and macOS.

- Build an **APRS position beacon** (lat/lon, symbol, comment), auto-appending live frequency/mode.
- **AX.25 UI frame** construction + **AFSK modulation** to audio (300-baud HF profile).
- Keys PTT, plays the packet audio through a chosen sound device, releases PTT — with the safety fixes from this session (guaranteed key-down, no cross-keying, callsign validated before transmit).
- **Manual** "Send Beacon" and **automatic** periodic beaconing at a configured interval.

13. Settings & platform

- One Settings window covering connection, APRS, callsign lookup, LoTW/TQSL, HRD, and N1MM.
- Cross-platform audio (NAudio/WASAPI on Windows, `afplay` on macOS).
- Settings persisted to JSON; secrets kept out of source control.
- **Windows, macOS, Linux desktop, and Raspberry Pi OS** all supported via Avalonia.

14. Architecture (for context)

Four clean layers: **CivEngine** (CI-V framing, serial I/O, APRS/AFSK) → **RigModel** (Transceiver + the network layer) → **Services** (all the integrations above) → **UI** (Avalonia views/view-models). 283 automated tests.

What it does NOT do yet — the brainstorm springboard

These are the natural places to take it next. Items marked (*documented*) already have notes in CLAUDE.md.

Small / near-term

- **Remote Power ON/OFF** (*documented*) — CI-V 0x18; one method + a button. Caveat: can only wake a radio that's in low-power standby, not fully off.
- **Real-hardware validation** of Phase 9 remote mode over an actual network link.
- **Meter scaling verification** — Po/ALC/scope-span byte layouts need confirming against a real radio (flagged in the audit).

Medium

- **IC-705 support** (*documented*) — add to the radio enum, verify meter scaling; optional GPS/D-PRS and D-STAR would be genuinely new feature work.
- **UI redesign** (*documented*) — you've said the current UI isn't satisfying; this is the gate for the comprehensive User Manual. Open questions: color/theme, transceiver-panel vs. flat dashboard, tabs vs. one scrolling window.
- **Phase 11: clickable radio front-panel** (*documented*) — a photo/vector of the rig with clickable controls.

Large / its own project

- **Phase 12: Remote Audio** (*documented*) — real-time low-latency audio capture + streaming + playback over IP, to fully match RS-BA1. Deliberately scoped as its own multi-session phase (needs a new capture interface, streaming protocol, jitter buffering).
- **Comprehensive User Manual** (*documented*) — real screenshots, click-by-click, every quirk; triggered once the UI-redesign question is resolved.

Not yet considered (open field for brainstorming)

- CW keyer / message memories; voice recording/playback (RS-BA1 has these).
- Rotator control, band/antenna switching, amplifier keying.
- Cluster / DX spot integration, band map.
- Digital-mode decode (FT8/RTTY) in-app vs. the current WSJT-X hand-off.
- More contests; general-logging niceties (QSL cards, statistics, maps).
- Mobile/tablet or web front-end to the headless server.